Assignement

2023-03-30

```
## tibble [53,940 x 10] (S3: tbl_df/tbl/data.frame)
##  $ carat  : num [1:53940] 0.23 0.21 0.23 0.29 0.31 0.24 0.24 0.26 0.22 0.23 ...
##  $ cut    : Ord.factor w/ 5 levels "Fair"<"Good"<..: 5 4 2 4 2 3 3 3 1 3 ...
##  $ color  : Ord.factor w/ 7 levels "D"<"E"<"F"<"G"<..: 2 2 2 6 7 7 6 5 2 5 ...
##  $ clarity: Ord.factor w/ 8 levels "I1"<"SI2"<"SI1"<..: 2 3 5 4 2 6 7 3 4 5 ...
##  $ depth  : num [1:53940] 61.5 59.8 56.9 62.4 63.3 62.8 62.3 61.9 65.1 59.4 ...
##  $ table  : num [1:53940] 55 61 65 58 58 57 57 55 61 61 ...
##  $ price  : int [1:53940] 326 326 327 334 335 336 336 337 337 338 ...
##  $ x      : num [1:53940] 3.95 3.89 4.05 4.2 4.34 3.94 3.95 4.07 3.87 4 ...
##  $ y      : num [1:53940] 3.98 3.84 4.07 4.23 4.35 3.96 3.98 4.11 3.78 4.05 ...
##  $ z      : num [1:53940] 2.43 2.31 2.31 2.63 2.75 2.48 2.47 2.53 2.49 2.39 ...
```

Question 1

Assumptions

1. The data is made up of basic random values drawn from the population
2. First, we test the hypothesis of normality of the dependent variable therefore the price of diamonds

```
##
##  Shapiro-Wilk normality test
##
## data:  diamonds$price[1:5000]
## W = 0.66175, p-value < 2.2e-16
```

Since the p-value $<$ 2.2e-16, implies that data distribution don't follows the normal distribution.

3. Next, we will test the hypothesis of equality of variances between the groups.

```
##
##  Bartlett test of homogeneity of variances
##
## data:  diamonds$price by diamonds$cut
## Bartlett's K-squared = 406.7, df = 4, p-value < 2.2e-16
```

Since the p-value$<$ 2.2e-16, implies that the variances don't to be equal.

ANOVA

Let m1, m2, m3, m4 and m5 respectively be the mean prices of diamonds which group are "Ideal", "Premium", "Good", "Very Good", and "Fair".

H0: m1=m2=m3=m4=m5 Ha: At least one mean different

```
##
##  One-way analysis of means (not assuming equal variances)
##
## data:  price and cut
## F = 166.04, num df = 4.0, denom df = 9398.6, p-value < 2.2e-16
```

Conclusion

With an alpha of 0.05, and a p-value< 2.2e-16, we reject the null and stat that we have sufficient evidence to show that there was a effect of cut on diamond prices.

Question 2

The hypotheses are:

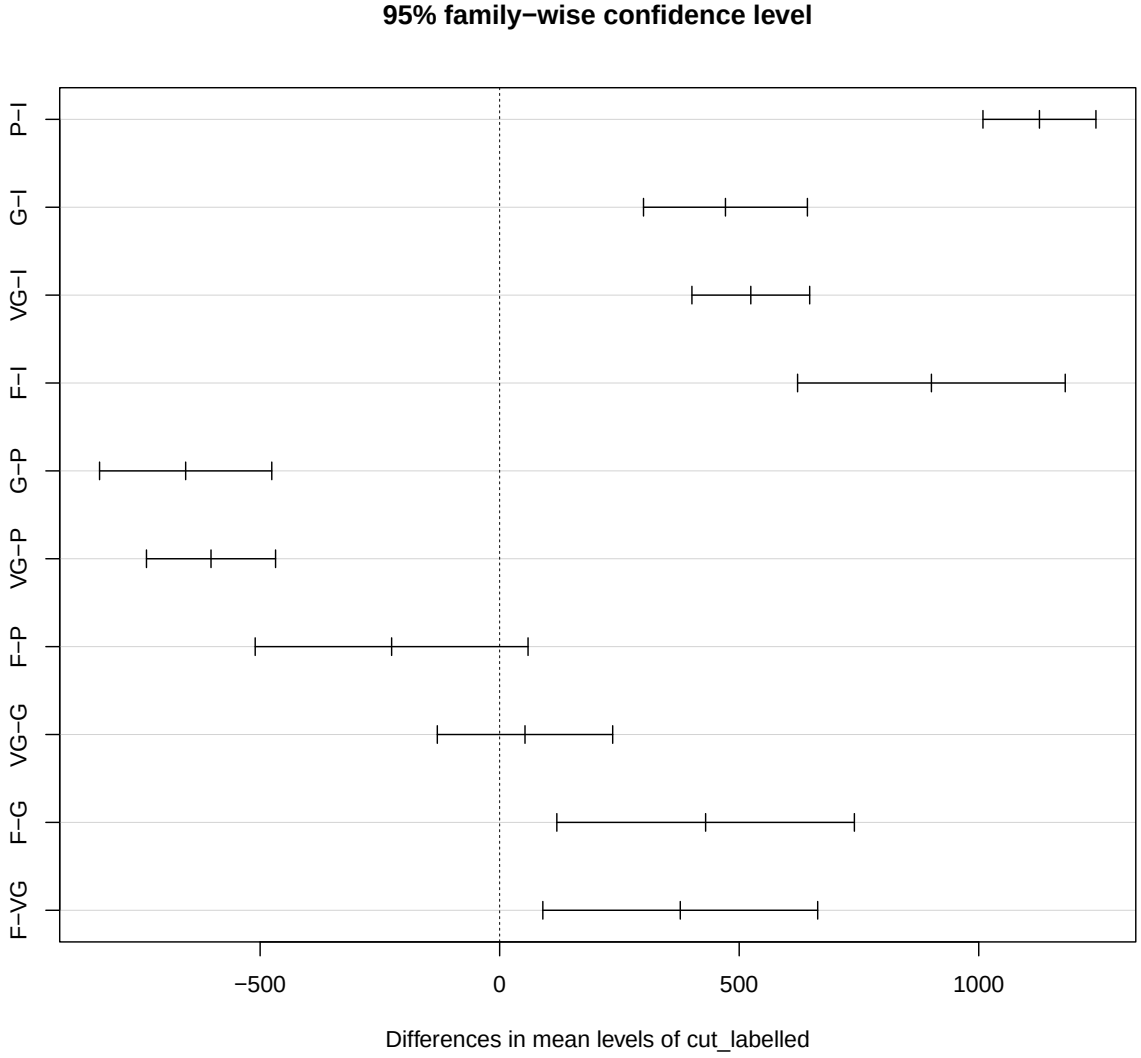
- H0: m1=m2=m3=m4=m5
- Ha: At least one mean different

```
##
##  One-way analysis of means
##
## data:  price and cut
## F = 175.69, num df = 4, denom df = 53935, p-value < 2.2e-16
```

There is no obvious difference in p-value but the F-statistic is different as well as "num df" and "denom df".

Question 3

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = price ~ cut_labelled, data = diamonds)
##
## $cut_labelled
##            diff       lwr        upr     p adj
## P-I  1126.71573 1008.8088 1244.62267 0.0000000
## G-I   471.32248  300.2823  642.36268 0.0000000
## VG-I  524.21792  401.3312  647.10467 0.0000000
## F-I   901.21579  621.8602 1180.57139 0.0000000
## G-P  -655.39325 -835.1353 -475.65120 0.0000000
## VG-P -602.49781 -737.2331 -467.76249 0.0000000
## F-P  -225.49994 -510.2665   59.26664 0.1950425
## VG-G   52.89544 -130.1519  235.94273 0.9341158
## F-G   429.89331  119.3378  740.44880 0.0014980
## F-VG  376.99787   90.1336  663.86215 0.0031094
```



Interpretations

The average price of diamonds in the "Premium", "Good", "Very Good", "Fair" categories is significantly higher than that of the "Ideal" category (with estimated difference of 1126.71573, 471.32248, 524.21792, 901.21579 respectively and p-value of 0.0 for all)

The average price of diamonds in the "Premium" category is significantly higher than in the "Good" and "Very Good" categories (with estimated difference of 655.39325, 602.49781 respectively and p-value of 0.0 for all).

The average price of diamonds in the "Fair" category is significantly higher than in the "Good" and "Very Good" categories (with estimated difference of 429.89331, 376.99787 respectively and p-value of 0.0014980 and 0.0031094 respectively.

Question 4

The hypotheses are:

- H0: m3=m4

- Ha: m3!=m4

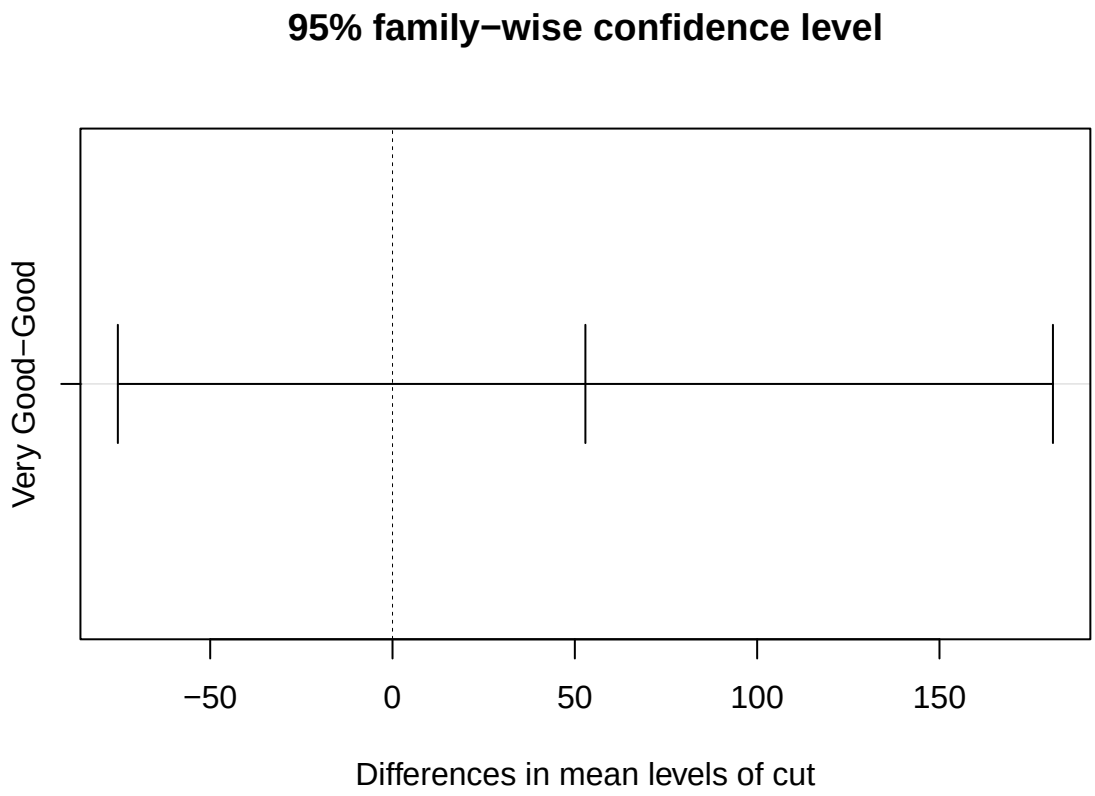
```
##
##  One-way analysis of means
##
## data:  price and cut
## F = 0.65381, num df = 1, denom df = 16986, p-value = 0.4188
```

Conclusion

With an alpha of 0.05, and a p-value of 0.4188, we fail to reject the null and stat that we have insufficient evidence to show that there was a cut effect of on diamond prices.

Question 5

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = price ~ cut, data = df)
##
## $cut
##                     diff       lwr      upr     p adj
## Very Good-Good 52.89544 -75.32923 181.1201 0.4187652
```



As the p-value is not significant (p-value=0.41>0.05) we cannot give an exact conclusion.

Question 6

```
##
## Call:
## lm(formula = price ~ cut + color + carat + depth, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17265.2   -751.5    -83.6    544.2  12256.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -910.028    286.893  -3.172  0.00151 **
## cut.L        1177.948     26.101  45.130  < 2e-16 ***
## cut.Q        -491.180     22.520 -21.811  < 2e-16 ***
## cut.C         370.862     19.151  19.365  < 2e-16 ***
## cut^4          85.419     15.425   5.538 3.08e-08 ***
## color.L     -1572.449     21.727 -72.373  < 2e-16 ***
## color.Q      -732.336     19.849 -36.896  < 2e-16 ***
## color.C      -109.973     18.631  -5.903 3.60e-09 ***
## color^4        82.481     17.106   4.822 1.43e-06 ***
## color^5      -137.643     16.174  -8.510  < 2e-16 ***
## color^6      -161.317     14.672 -10.995  < 2e-16 ***
## carat        8183.774     13.889 589.216  < 2e-16 ***
## depth         -35.979      4.602  -7.819 5.43e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1431 on 53927 degrees of freedom
## Multiple R-squared:  0.8713, Adjusted R-squared:  0.8713
## F-statistic: 3.042e+04 on 12 and 53927 DF,  p-value: < 2.2e-16
```

The coefficient of determination is 0.8713, which indicates that the model accounts for 87.13% of the variation in the diamond price. - An increase of one unit of depth results in a decrease of 35.979 in the diamond - An increase of one unit of carat results in a increase of 8183.774 in the diamond price. - If the item is in the "Ideal" category, there is an increase of 1177.948 in the diamond price. A decrease of 491.180 for "Premium", an increase of 370.82 for "Good", an increase of 85.419 for "Very Good".

Question 7

```
## Start:  AIC=758426.5
## price ~ carat + cut + color + clarity + depth + table + x + y +
##     z
```

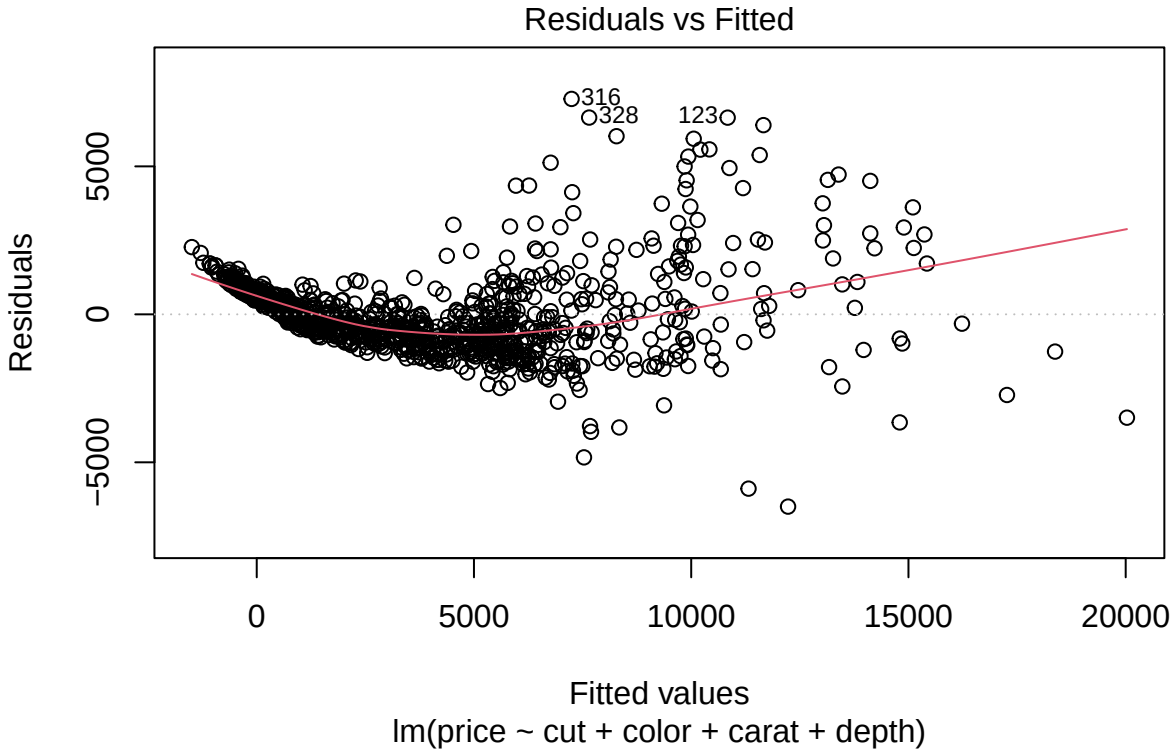
```
##
## Call:
## lm(formula = price ~ carat + cut + color + clarity + depth +
##     table + x + y + z, data = subset(diamonds, select = -c(cut_labelled)))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21376.0   -592.4   -183.5    376.4  10694.2
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
```

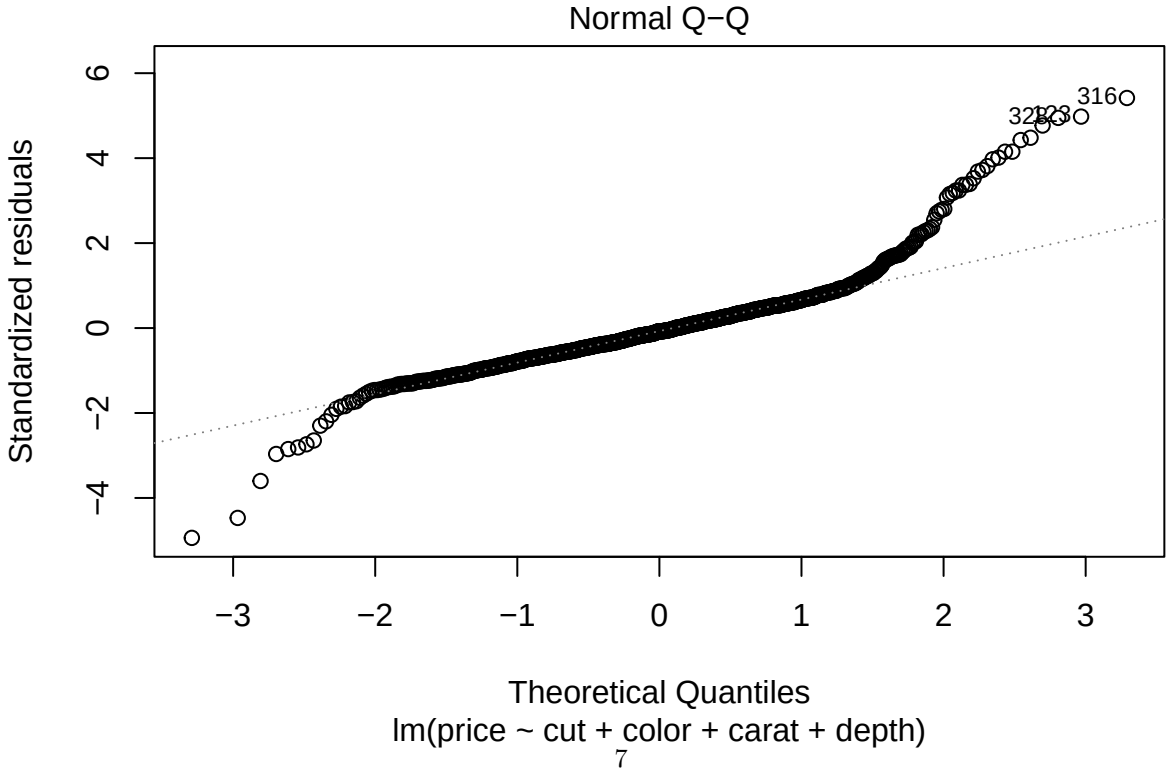
```
## (Intercept)  5753.762    396.630   14.507  < 2e-16 ***
## carat       11256.978     48.628  231.494  < 2e-16 ***
## cut.L         584.457     22.478   26.001  < 2e-16 ***
## cut.Q        -301.908     17.994  -16.778  < 2e-16 ***
## cut.C         148.035     15.483    9.561  < 2e-16 ***
## cut^4         -20.794     12.377   -1.680  0.09294 .
## color.L     -1952.160     17.342 -112.570  < 2e-16 ***
## color.Q      -672.054     15.777  -42.597  < 2e-16 ***
## color.C      -165.283     14.725  -11.225  < 2e-16 ***
## color^4        38.195     13.527    2.824  0.00475 **
## color^5       -95.793     12.776   -7.498 6.59e-14 ***
## color^6       -48.466     11.614   -4.173 3.01e-05 ***
## clarity.L    4097.431     30.259  135.414  < 2e-16 ***
## clarity.Q   -1925.004     28.227  -68.197  < 2e-16 ***
## clarity.C     982.205     24.152   40.668  < 2e-16 ***
## clarity^4    -364.918     19.285  -18.922  < 2e-16 ***
## clarity^5     233.563     15.752   14.828  < 2e-16 ***
## clarity^6       6.883     13.715    0.502  0.61575
## clarity^7      90.640     12.103    7.489 7.06e-14 ***
## depth         -63.806      4.535  -14.071  < 2e-16 ***
## table         -26.474      2.912   -9.092  < 2e-16 ***
## x           -1008.261     32.898  -30.648  < 2e-16 ***
## y               9.609     19.333    0.497  0.61918
## z             -50.119     33.486   -1.497  0.13448
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1130 on 53916 degrees of freedom
## Multiple R-squared:  0.9198, Adjusted R-squared:  0.9198
## F-statistic: 2.688e+04 on 23 and 53916 DF,  p-value: < 2.2e-16
```

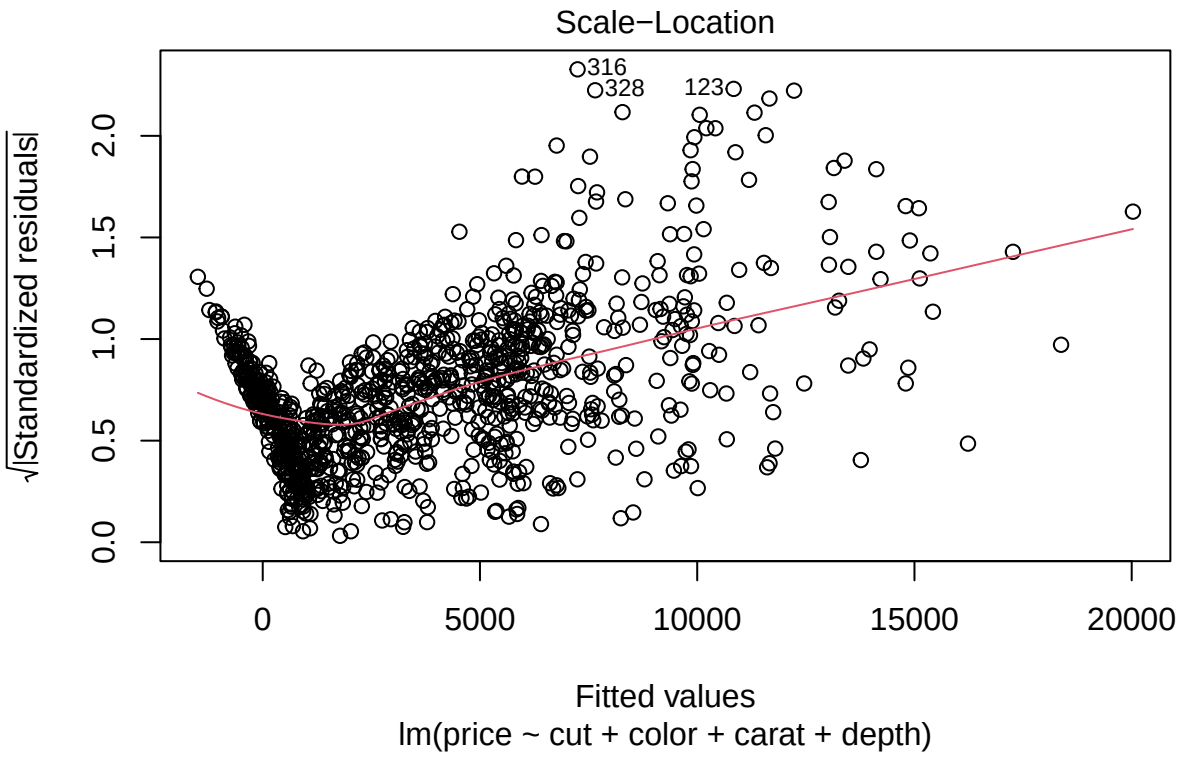
Conclusion

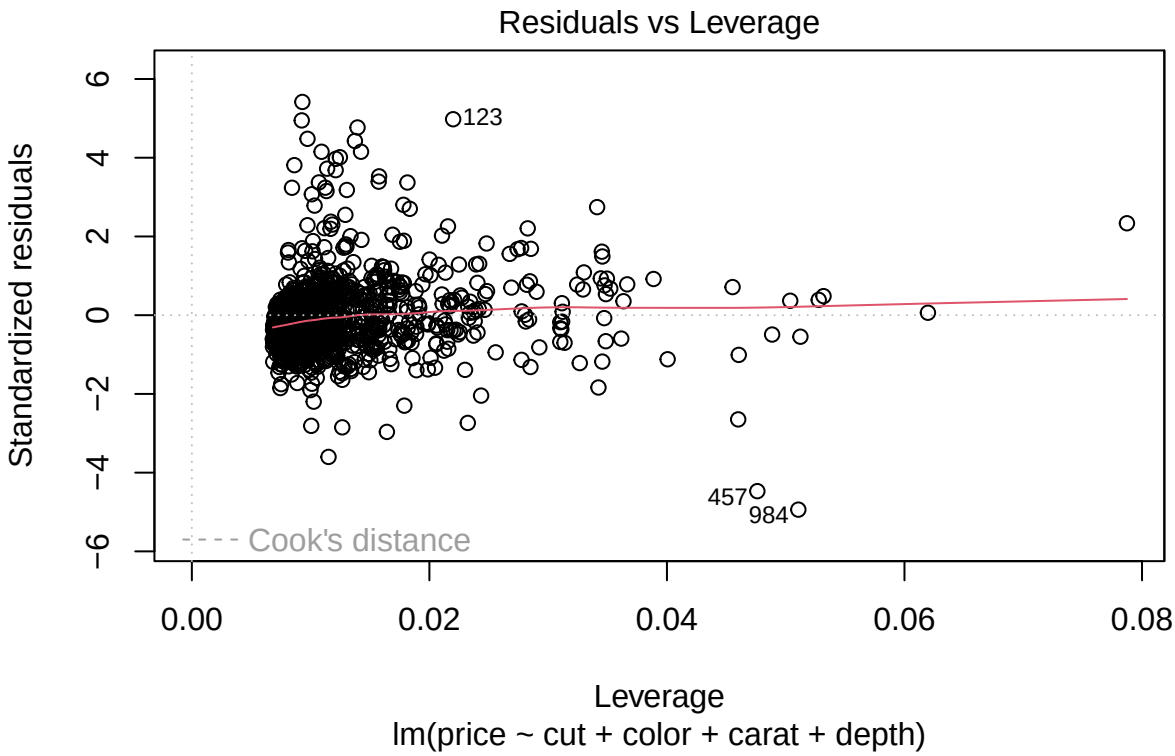
The determination coefficient of the R-squared model is 0.9198. This indicates that the model expresses 91.98% of the variation in diamond prices.

Question 8


Residuals vs Fitted
316
328
123
Residuals
5000
0
−5000
0
5000
10000
15000
20000
Fitted values
lm(price ~ cut + color + carat + depth)


Normal Q−Q
316
Standardized residuals
6
4
2
0
−2
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(price ~ cut + color + carat + depth)


Scale−Location
316
328
123
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
0
5000
10000
15000
20000
Fitted values
lm(price ~ cut + color + carat + depth)



Interpretation

There is a predefined light parabolic shape on one side of the Residuals vs Fitted graph. This tells us that the model is incomplete: it is missing a quadratic factor that could explain a larger variation in diamond prices. The normal Q - Q plot has enough points off the line, this indicates that the residuals do not follow a normal distribution. The points 785 and 905 stand out on each of the plots, this indicates that there is a problem with these observations.